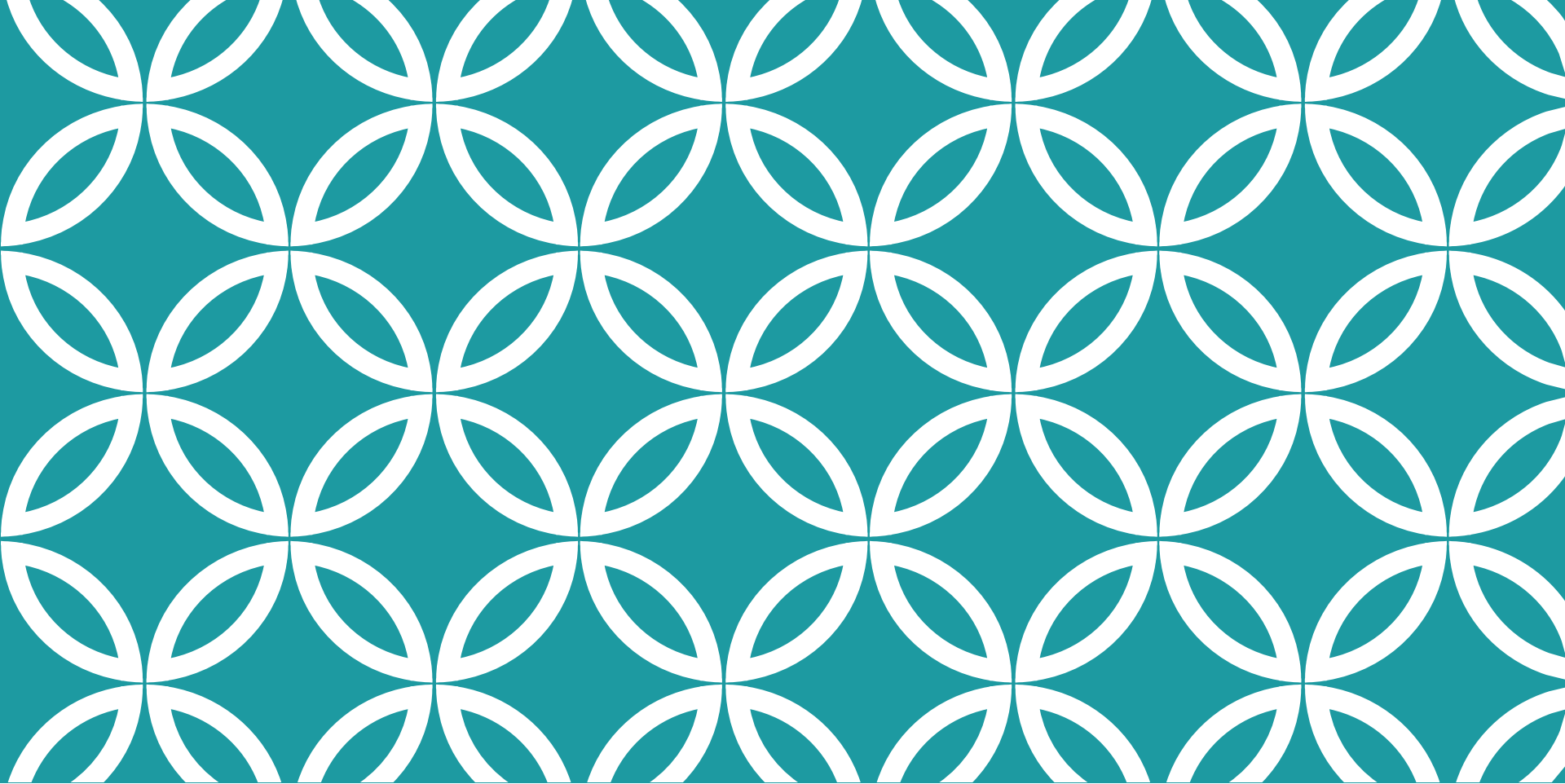


POLE-ZERO MODELING

Reference Text Books:

1. Fundamentals of Speech recognition – L. Rabiner and B. Juang, Prentice Hall signal processing series.
2. Discrete-time speech signal processing: principles and practice, Thomas F. Quatieri, Coth.



Z-TRANSFORM

Fourier Transform not
enough?

Stability of Systems

Z-transform

Z-domain

FOURIER-TRANSFORM NOT ENOUGH?

Fourier Transform/FFT is not suited for

1. Low-latency requirements
2. Constrained devices
3. IIR filters incorporating feedback

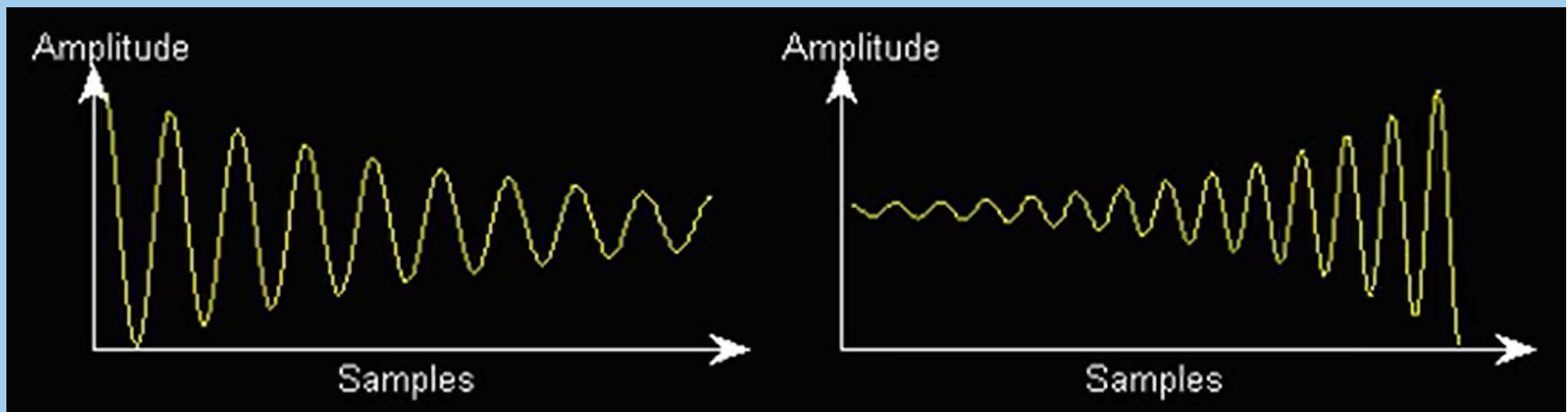
FOURIER-TRANSFORM NOT ENOUGH?

Understanding only the frequency of the signal is sometimes not enough.

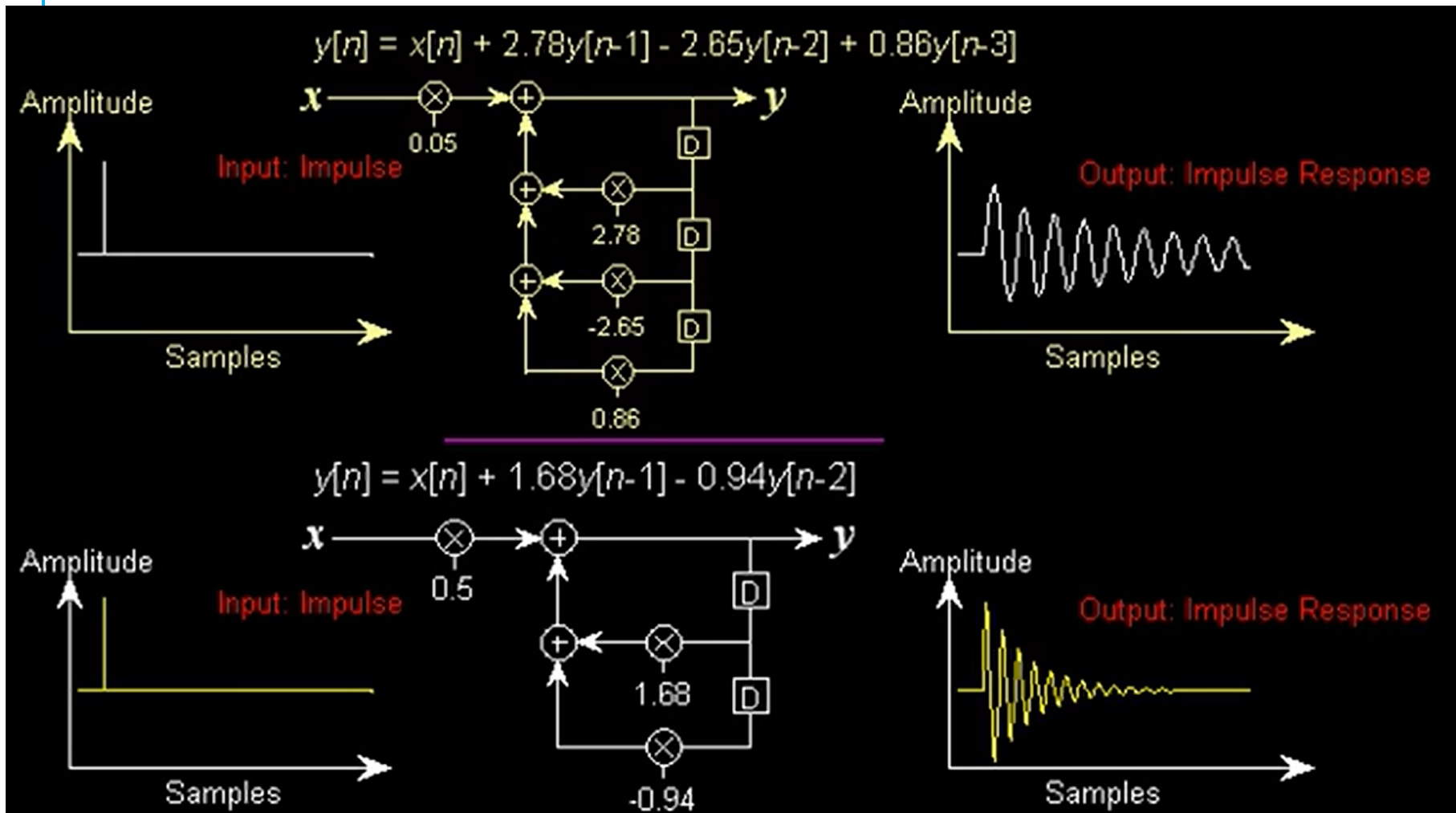
To know how the system is generating the output signal, we need information about the system as well.

Z-transform helps predict what will be the output of the system given an input and enables us to manipulate the system to get the desired output.

Z-transform identifies the presence of exponentially increasing or decreasing oscillations in the signal $x[n]$. – whether the system is **stable or unstable**.

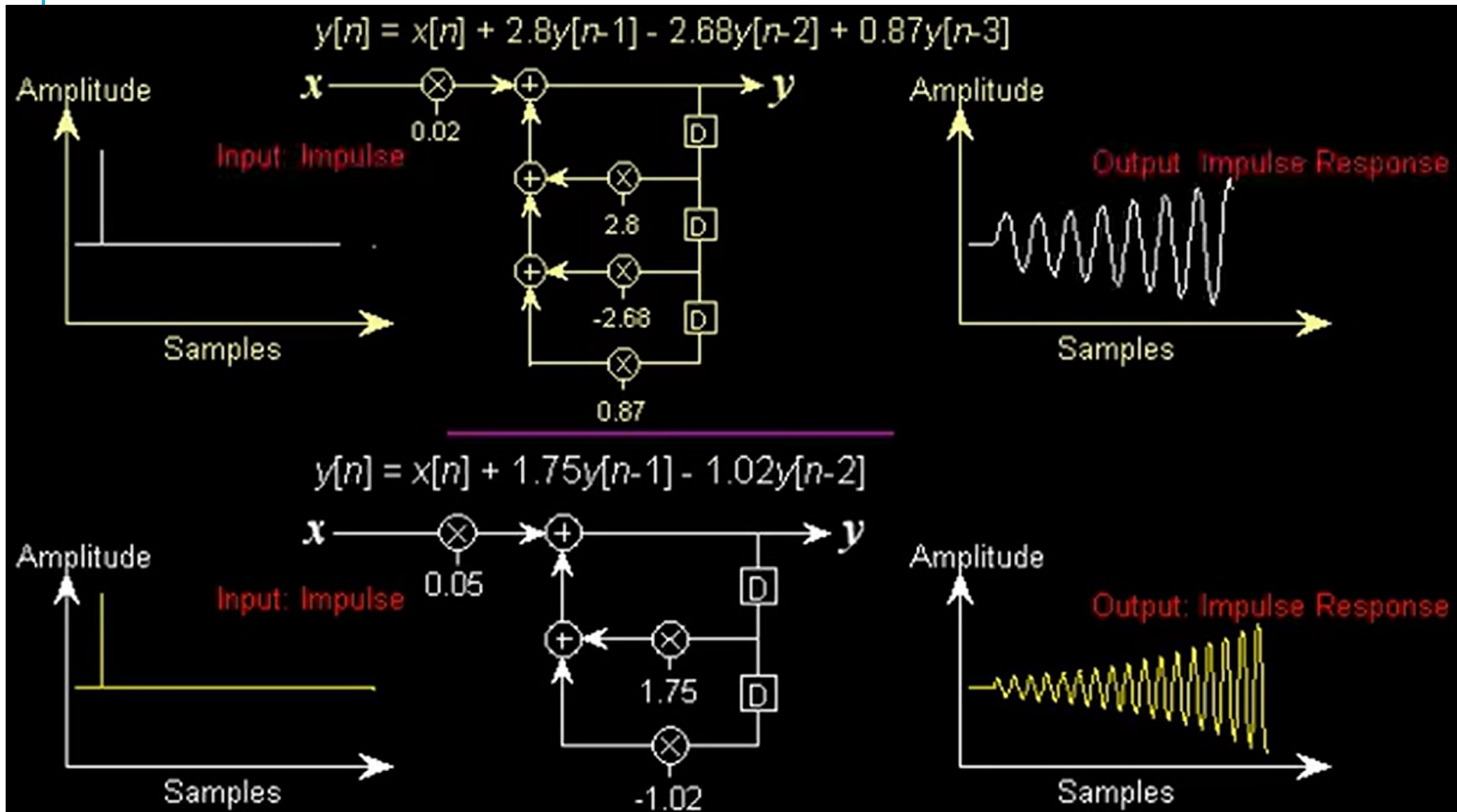


STABLE SYSTEMS

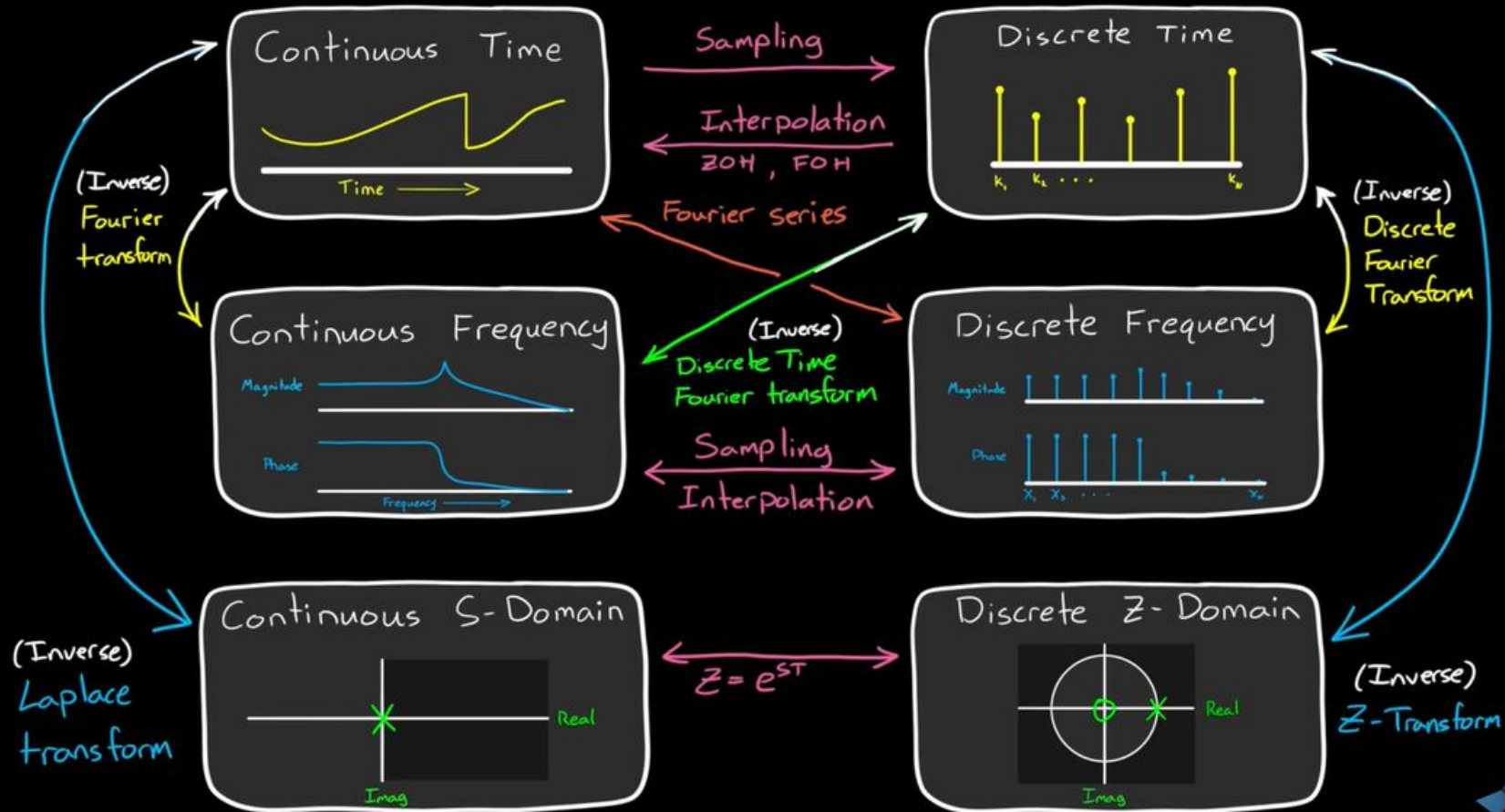


UNSTABLE SYSTEMS

Unstable systems not desirable



S-DOMAIN AND Z-DOMAIN OF SIGNAL



S-DOMAIN REPRESENTATION

The s-domain transforms difficult **time-domain differential equations** into **easy-to-solve algebraic equations**.

The s-domain (also known as the Laplace domain) is a complex frequency domain.

Fourier Transform maps signals to a purely imaginary frequency (ω).

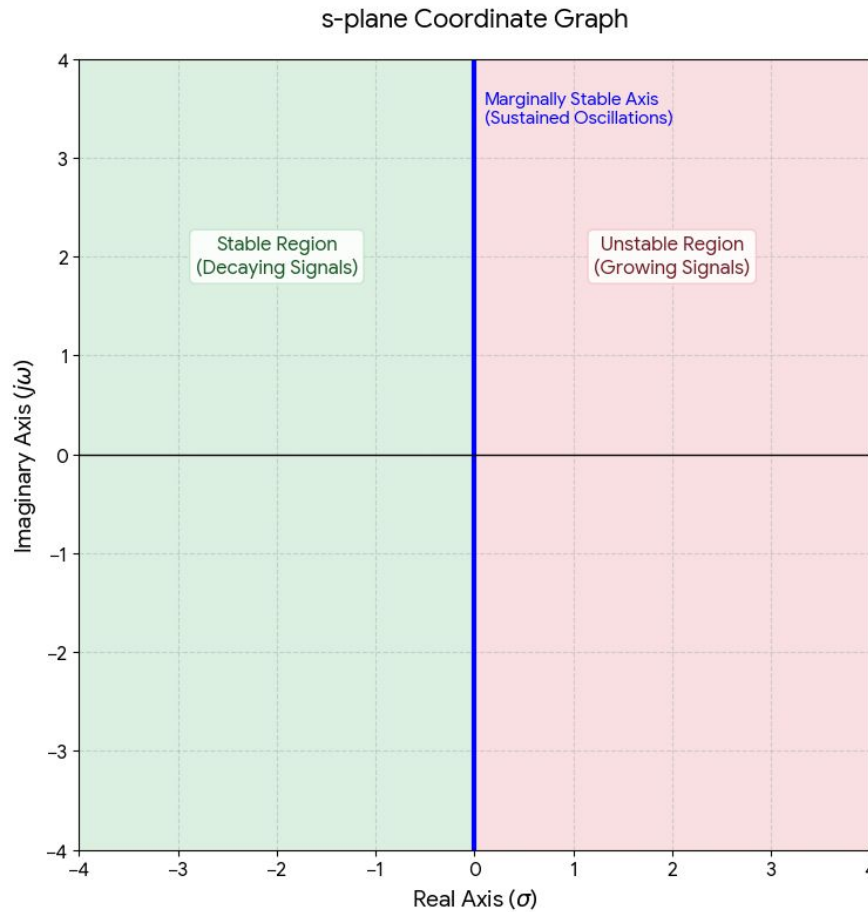
Laplace Transform maps time-domain signals to a complex variable (s):

$$s = \sigma + j\omega$$

where σ =damping rate, ω =angular frequency

S-DOMAIN REPRESENTATION

S-plane



S-DOMAIN REPRESENTATION

Comparison of continuous domains:

Feature	Time Domain (t)	Fourier Domain ($j\omega$)	Laplace s-Domain ($s = \sigma + j\omega$)
Primary Math	Differential Equations	Algebraic Equations	Algebraic Equations
Signal Types	Real-world signals	Pure sinusoids (steady-state)	Growing, decaying, or steady sinusoids
Initial Conditions	Difficult to plug in	Assumed to be zero	Built directly into formulas
System Insights	Hard to visualize stability	Excellent for filtering/ audio	Best for control systems & stability

Z-DOMAIN OF SIGNAL

It is the mathematical equivalent of the s-domain built specifically for discrete-time (digital) signals and systems.

z-domain handles discrete-time systems via the Z-Transform.

The complex variable z is defined as:

$$z = e^{sT} = e^{(\sigma + j\omega)T} = e^{\sigma T} \cdot e^{j\omega T}$$

where T =sampling interval, $e^{\sigma T}$ =scaling term, $e^{j\omega T}$ =Trace a circle

While the s-domain maps stability to a left/right plane, the z-domain maps stability to the inside or outside of a Unit Circle ($|z| = 1$).

Z-TRANSFORM OF SIGNAL

Z is a complex value.

Discrete-time
Fourier transform

$$\sum_{n=-\infty}^{\infty} x[n] \cdot \underline{e^{-i\omega n} \cdot r^{-n}}$$

$$e^{-i\omega n} r^{-n}$$

$$(r \cdot e^{i\omega})^{-n}$$

$$z^{-n}$$

where

$$z = r e^{i\omega}$$

Makes it the
z-transform

Z-TRANSFORM OF SIGNAL

Z-transform converts a discrete-time signal into a z-domain or z-plane representation.

Z-transform analyze, design, and process discrete-time (digital) systems algebraically.

Let's say $x[n]$ is a time domain signal, sampled periodically.

For Z-transform, each sample is multiplied with z^{-n} .

$$X(z) = \mathcal{Z}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

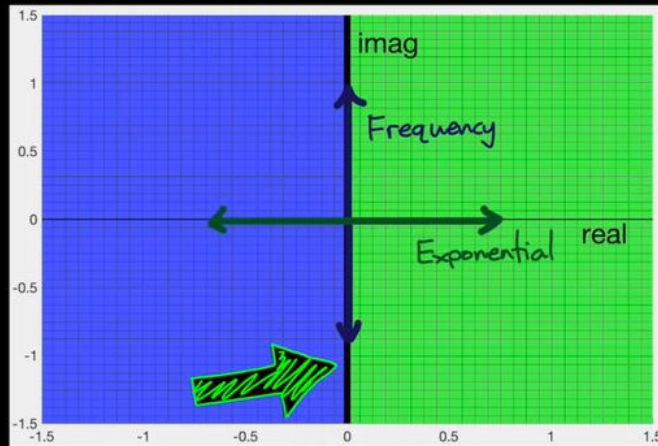
$x[n]$ time domain, over a variable n .

$X(z)$ in Z-domain, over a variable z .

Z-DOMAIN

S-PLANE TO Z-PLANE

S - Domain

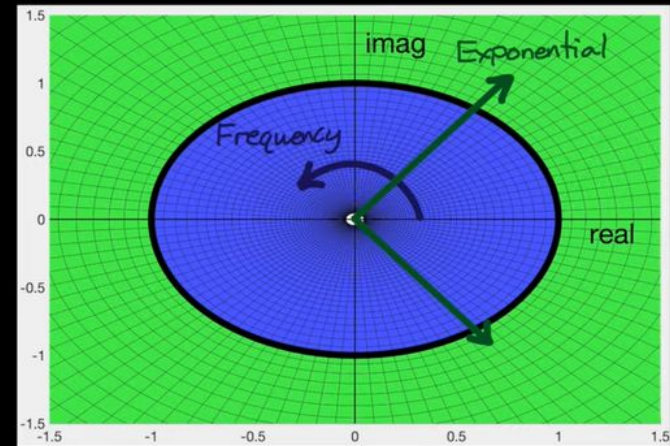


$$Z = e^{sT} \quad \text{where} \quad s = \sigma + j\omega$$

$$Z = (e^{\sigma T}) (e^{j\omega T})$$

Traces a circle
Scaling term

Z - Domain



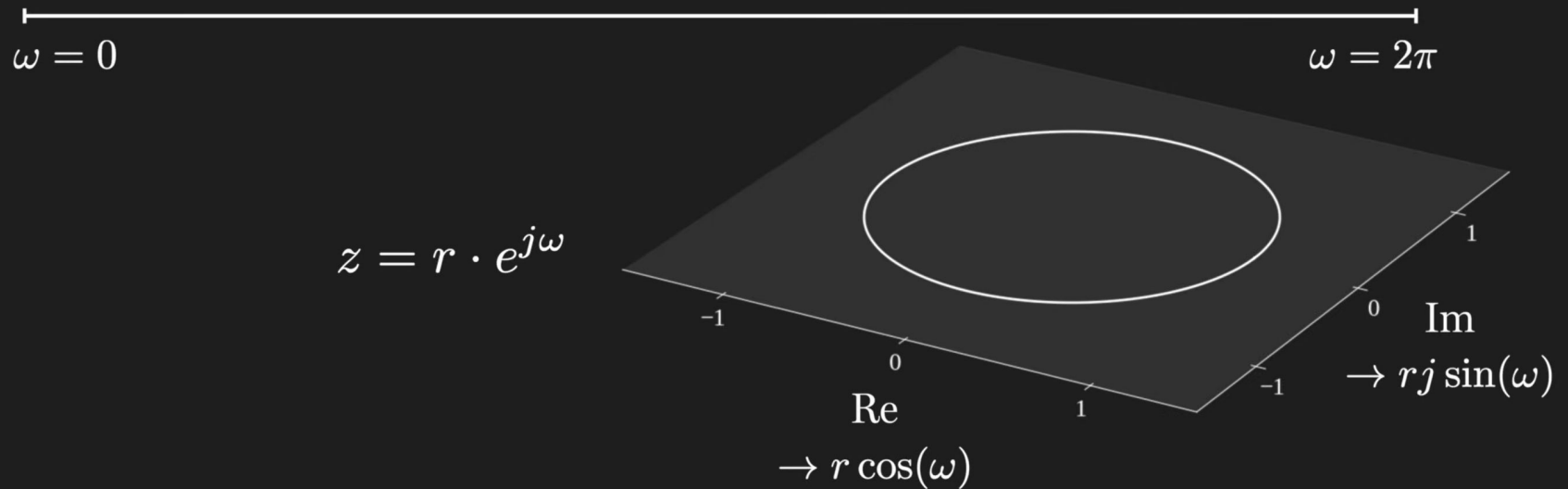
$$e^{j\omega T} = \cos(\omega T) + j\sin(\omega T)$$

$$e^{\sigma T} = \text{real number}$$

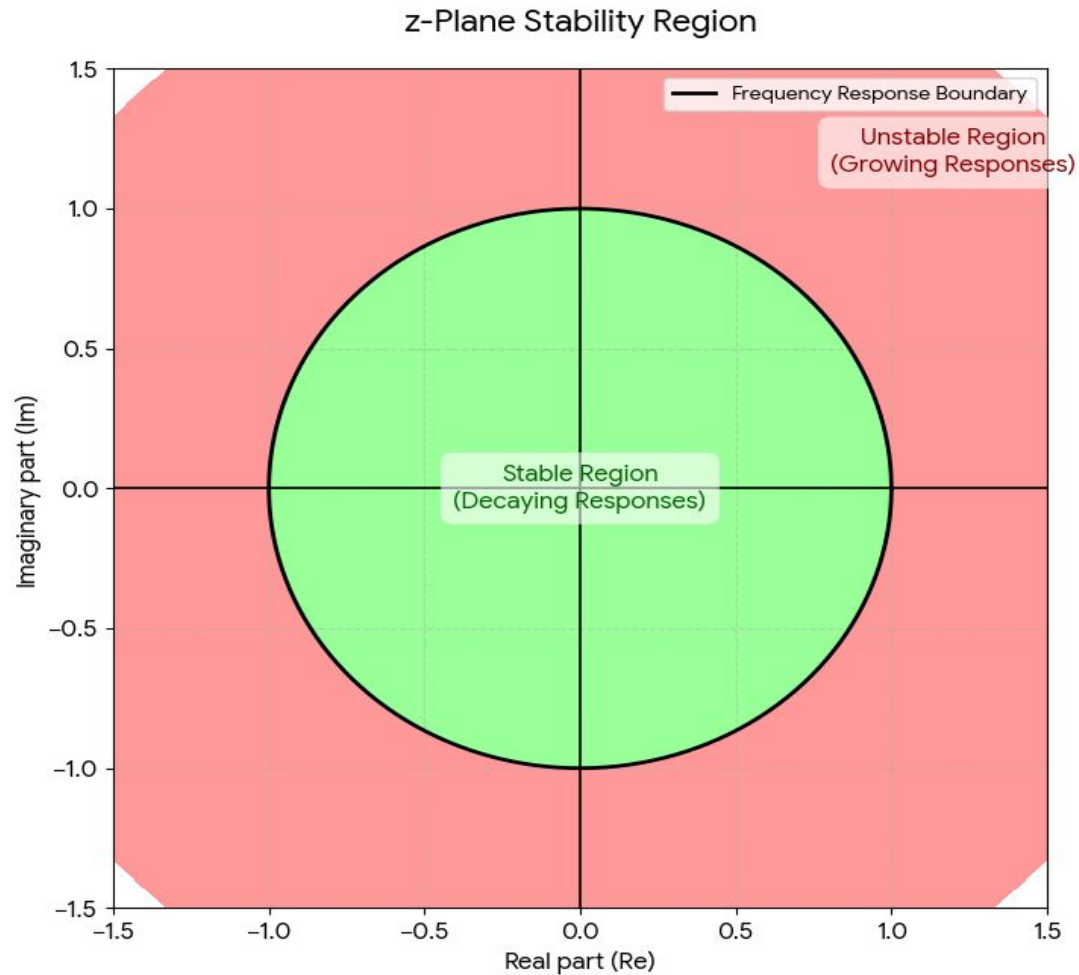


Z-DOMAIN

$$H(\omega) = \sum_{n=-\infty}^{+\infty} h[n] r \cdot e^{j\omega(-n)} \longrightarrow \sum_{n=-\infty}^{+\infty} h[n] z^{-n} = H(z)$$



Z-DOMAIN



Z-DOMAIN

$$x[n] \xleftrightarrow{z} X(z)$$

$$y[n] \xleftrightarrow{z} Y(z)$$

$$x[n] + y[n] \xleftrightarrow{z} X(z) + Y(z)$$

Linearity

$$x[n-1] \xleftrightarrow{z} X(z)z^{-1}$$

$$x[n-2] \xleftrightarrow{z} X(z)z^{-2}$$

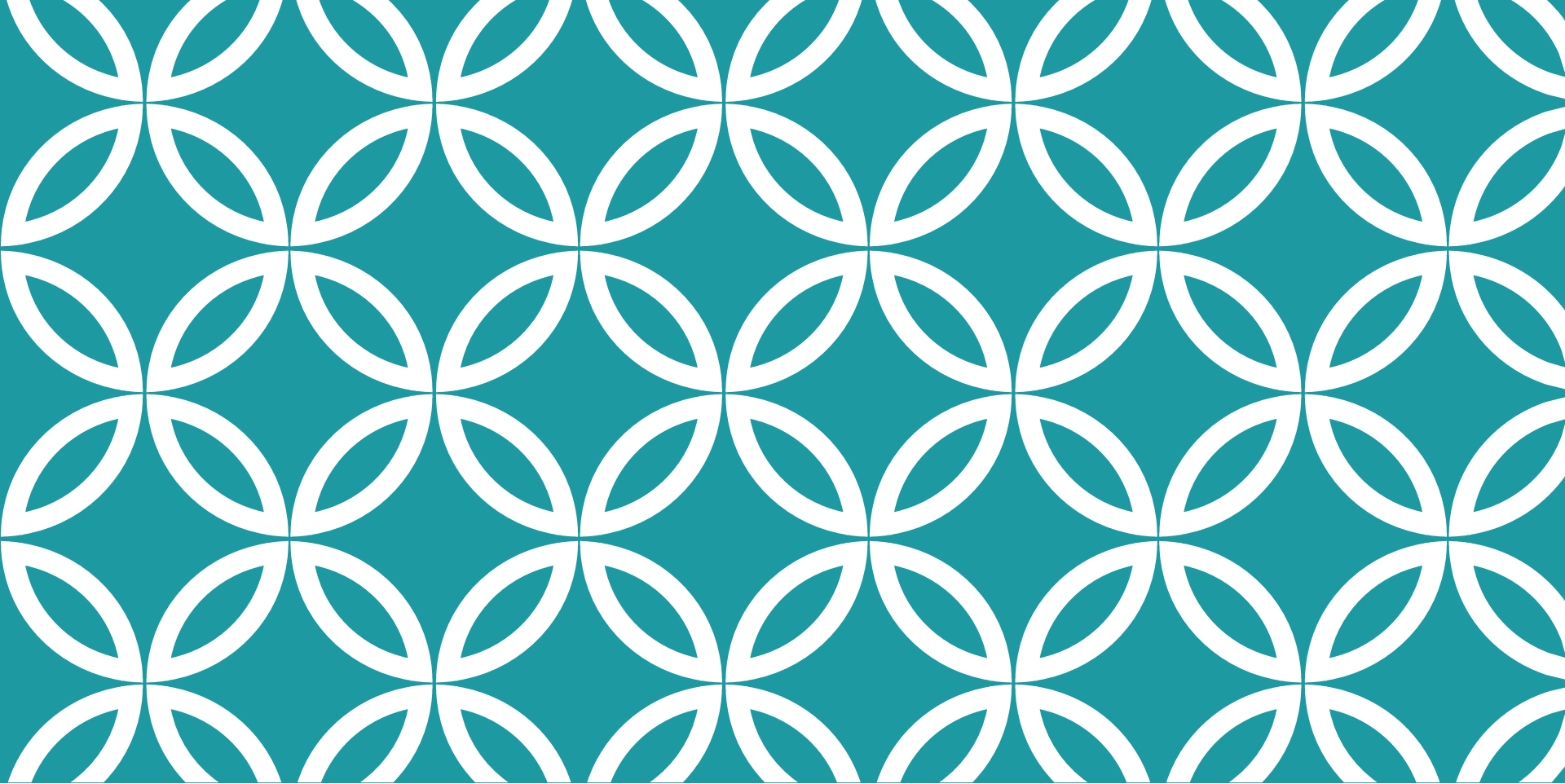
$$\vdots$$

$$x[n-k] \xleftrightarrow{z} X(z)z^{-k}$$

Shift
Property

$$H(z) = (1 - a_1 z^{-1})(1 - a_2 z^{-1})(1 - a_3 z^{-1}) \dots$$

Roots



POLE ZERO MODELING

Poles and Zeroes
Pole-Zero Plot
Pole-Zero Modeling for
Speech

TRANSFER FUNCTION IN Z-DOMAIN

A transfer function for a discrete-time LTI system is a rational function

It can be represented in Z-domain.

$$\begin{aligned} H(z) &= \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots + b_M z^{-M}}{a_0 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_M z^{-M}} \\ &= \frac{(z - z_1)(z - z_2) \dots}{(z - p_1)(z - p_2) \dots} \end{aligned}$$

$X(z)$ = Z-transform of the input signal

$Y(z)$ = Z-transform of the output signal

POLES AND ZEROES

$$H(z) = \frac{Y(z)}{X(z)} = \frac{(z-z_1)(z-z_2)\dots}{(z-p_1)(z-p_2)\dots}$$

$$X(z) = 0 \text{ for } z = p_1, p_2, \dots \quad Y(z) = 0 \text{ for } z = z_1, z_2, \dots$$

Here z_i are zeroes of the system and p_i are poles of the system.

Zeroes are **the roots of the numerator** for which transfer function becomes zero.

$$\lim_{z \rightarrow z_i} H(z) = 0.$$

Poles are the **roots of the denominator** for which transfer function becomes unbounded

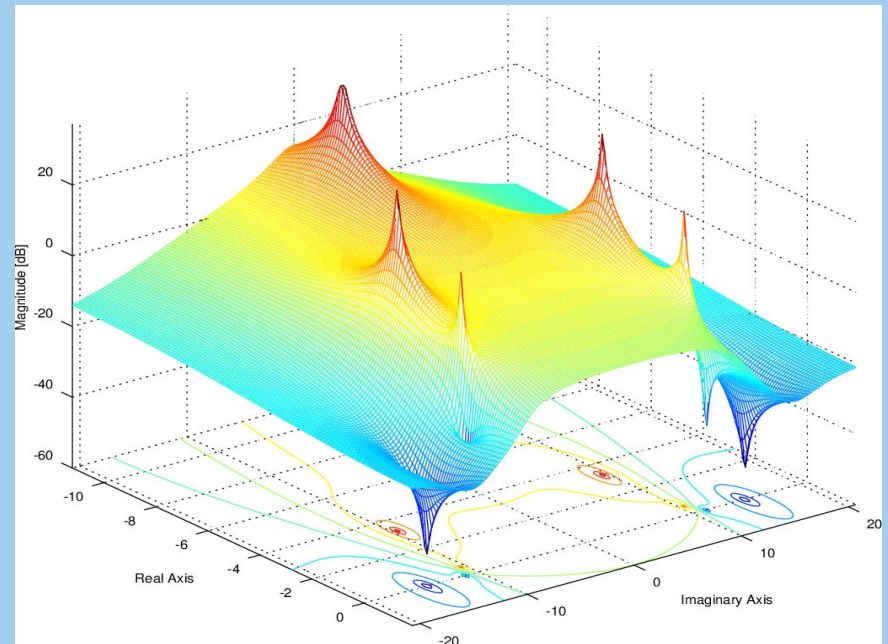
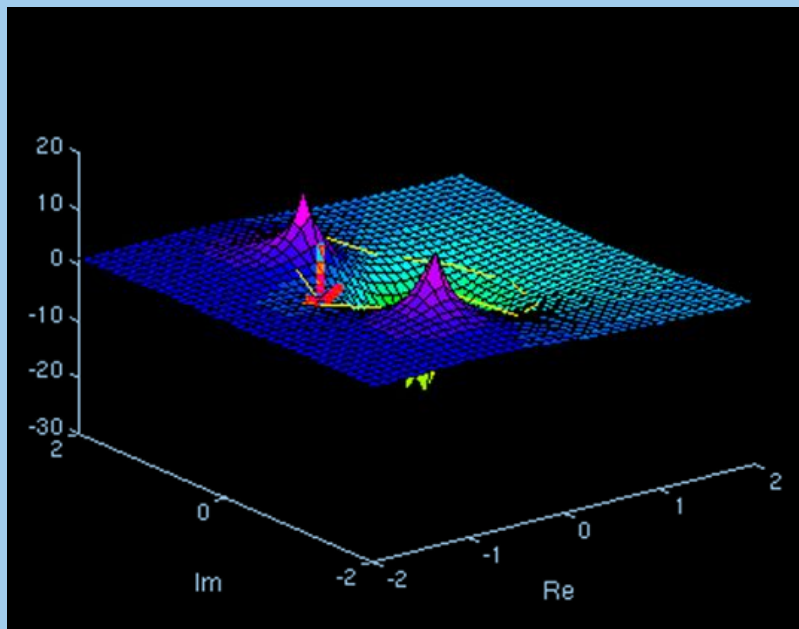
$$\lim_{z \rightarrow p_i} H(z) = \infty$$

All of the coefficients of polynomials $X(z)$ and $Y(z)$ are real, therefore the poles and zeros must be either purely real, or appear in complex conjugate pairs.

FREQUENCY RESPONSE OF POLES & ZEROES

Poles are also defined as “the frequency at which the system becomes infinite”. Hence the name pole. Cause **peaks** (resonance) at the frequency.

Zeros are the frequencies where the response is completely attenuated (cancelled). Cause **notches** (attenuation) at the frequency.



SIGNIFICANCE OF POLES AND ZEROS

The location of the poles and zeros provide qualitative insights into the response characteristics of a system.

Pole-zero plot shows **stability** and **resonance of the system**.

In z-plane, **poles** must lie inside the unit circle (discrete-time). This indicates bounded response and stable system. Poles lead to sharp peaks in frequency response (resonance).

Zeros represent frequencies where the system output becomes zero. The frequency response can be shaped by cancelling certain frequency components. Notch filter can be designed by placing zeros accordingly.

SIGNIFICANCE OF POLES AND ZEROS

The distance of poles and zeros from the unit circle determines how strong or weak these effects are.

Poles near the unit circle $\rightarrow$ cause sharp peaks in the frequency response (resonance).

Poles farther inside the circle $\rightarrow$ weaker resonance, smoother response.

Poles outside the unit circle $\rightarrow$ unstable system (response grows unbounded).

Zeros near the unit circle $\rightarrow$ create deep notches (strong attenuation) at those frequencies.

Zeros farther away $\rightarrow$ weaker attenuation, less noticeable effect.

Zeros exactly on the unit circle $\rightarrow$ complete cancellation at that frequency.

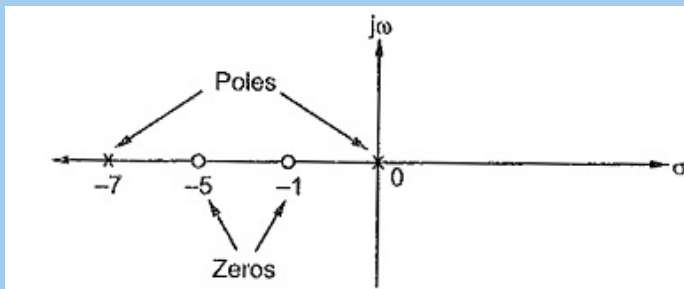
Poles and zeroes together shapes system's filtering behaviour. Pole-zero placement is the **design backbone of digital filters**.

POLE-ZERO PLOT

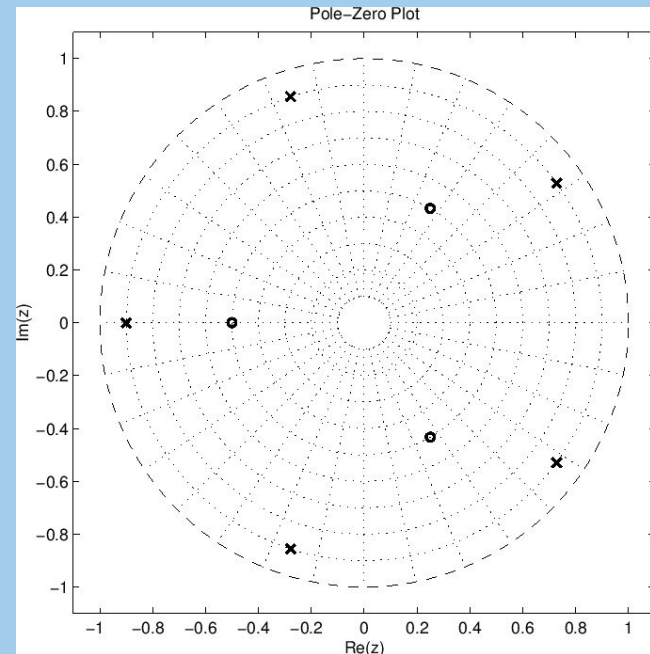
A pole-zero plot is plotted in the plane of a complex frequency domain, showing the location of the poles and zeros of the transfer function.

By convention, the poles of the system are indicated in the plot by an X while the zeros are indicated by a circle or O.

Poles and zeros must be either purely real, or appear in complex conjugate pairs.



s-plane pole-zero plot for continuous time-domain



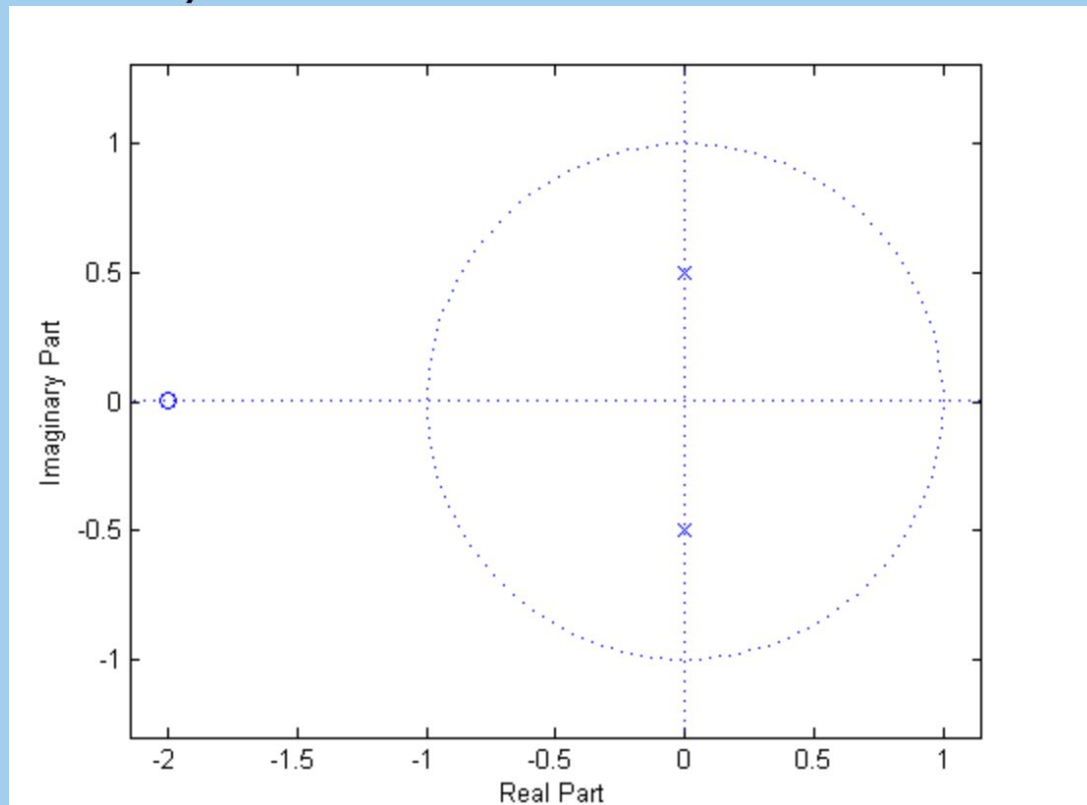
z-plane pole-zero plot for discrete time-domain

EXAMPLE

Let's consider the following second-order transfer function:

$$H(z) = \frac{z + 2}{z^2 + \frac{1}{4}}$$

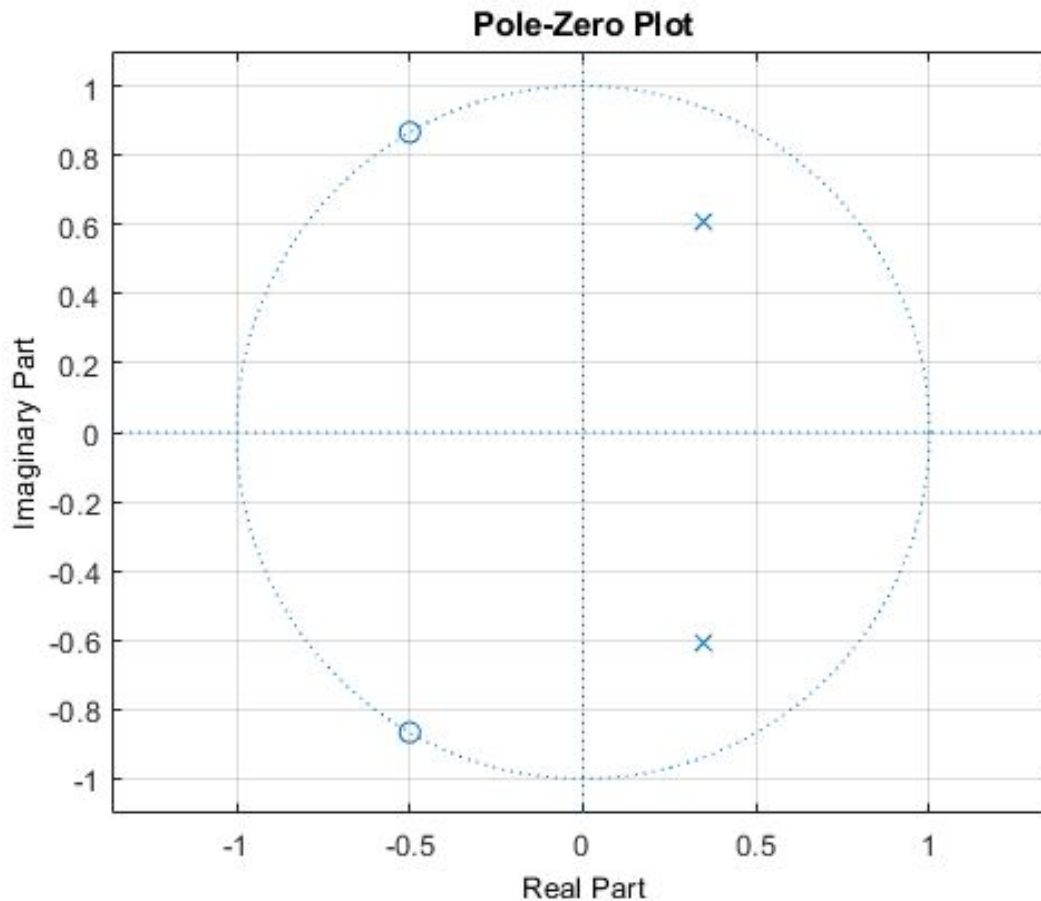
The pole-zero plot of the system is



EXAMPLE

pole-zero plot of the following second-order transfer function:

$$H(z) = \frac{(z + 0.5 + j0.82)(z + 0.5 - j0.82)}{(z - 0.35 + j0.6)(z - 0.35 - j0.6)}$$



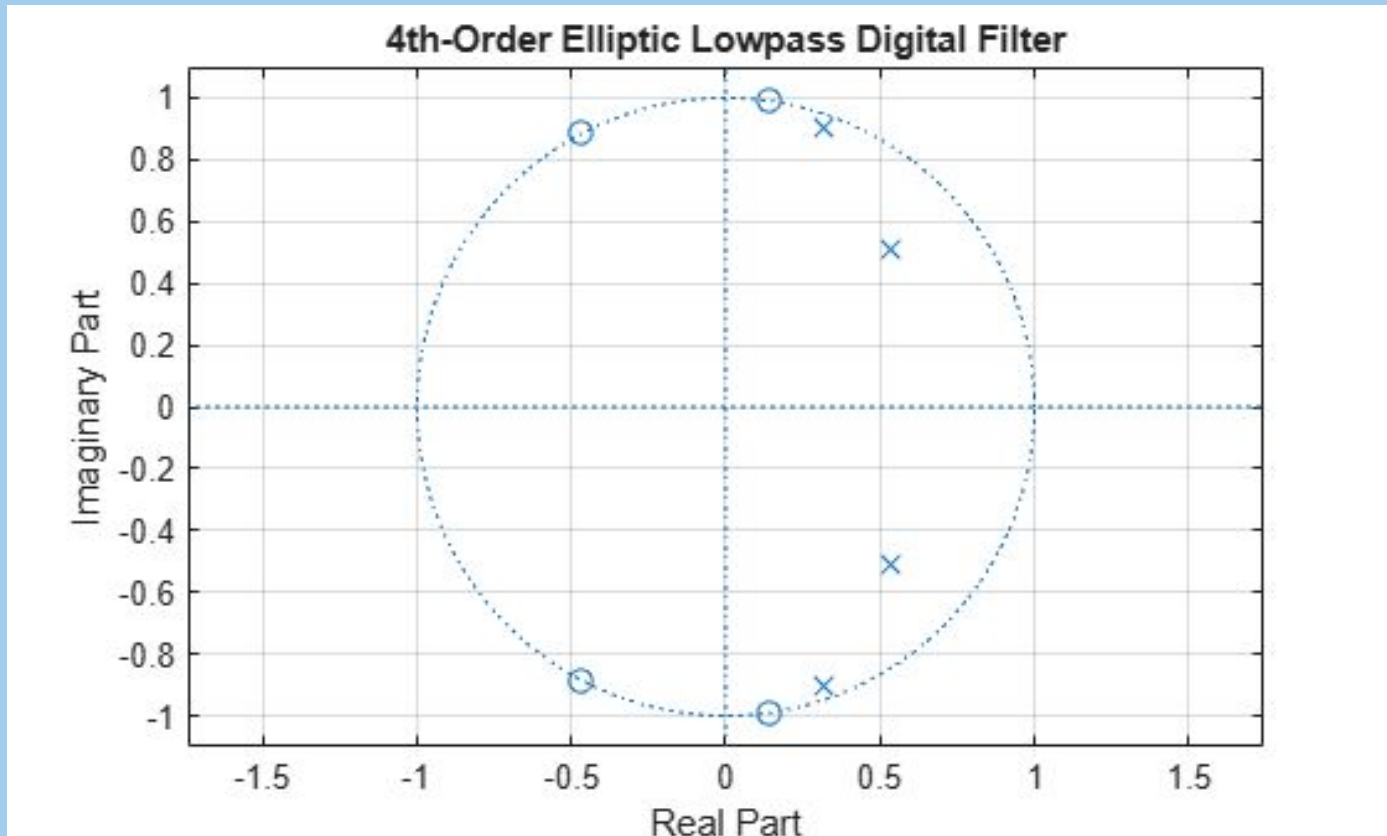
EXERCISE

Draw the pole-zero plot for the following third-order system

$$H(z) = \frac{3(z + 2)}{(z + 1)(z + 1 - 2j)(z - 1 + 2j)}$$

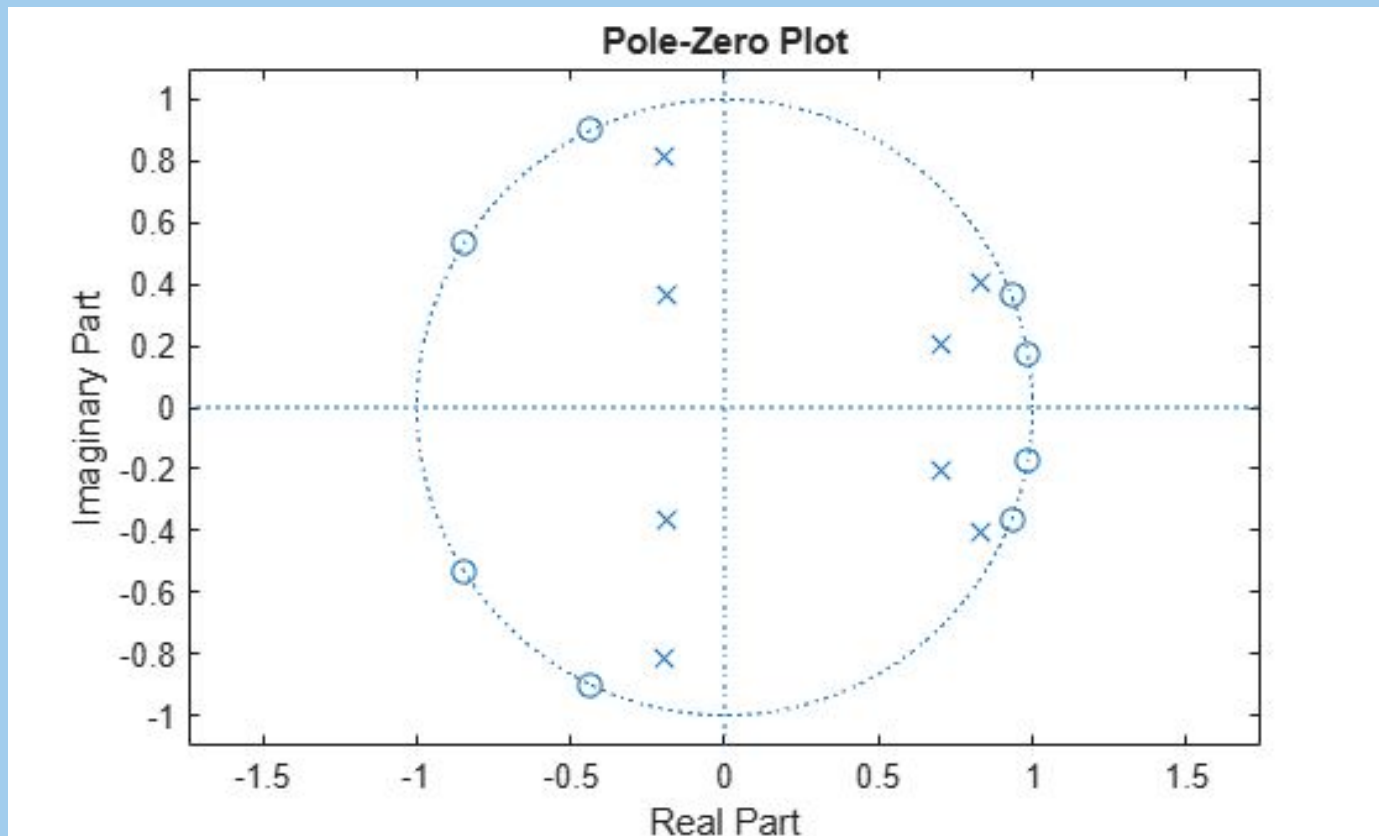
EXAMPLE

Pole-zero plotting of a 4th order low-pass digital filter



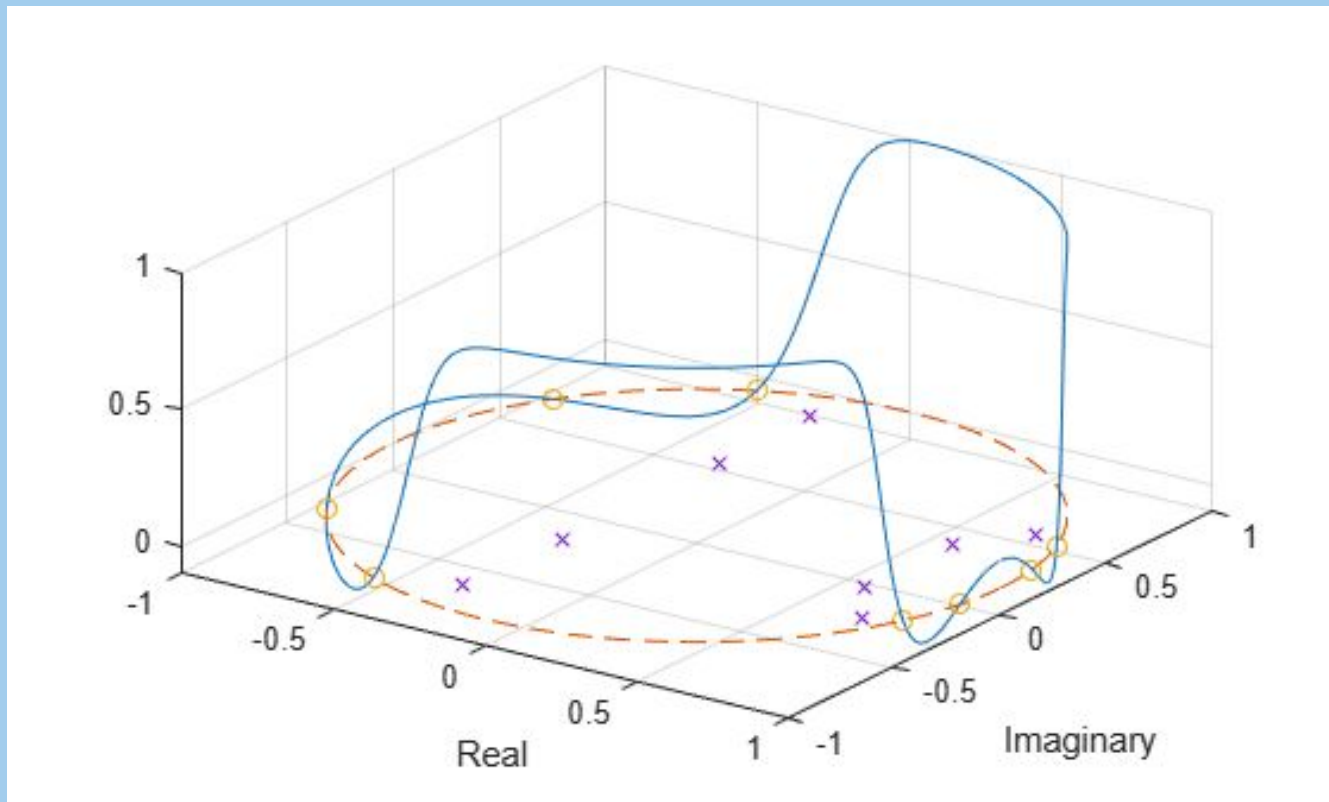
EXAMPLE

Pole-zero plotting of an 8th order low-pass digital filter



EXAMPLE

Frequency Response and Pole-zero plotting of an 8th order low-pass digital filter



FIR FILTER DESIGN USING POLE-ZERO

FIR filters have a **finite impulse response** — the output depends only on a finite number of past input samples.

Because there is no feedback, the denominator in the general filter equation is simply 1.

Transfer function for FIR filter:

$$H(z) = b_0 + b_1z^{-1} + b_2z^{-2} + \cdots + b_Nz^{-N}$$

This is a polynomial in z^{-1} with **only zeros** (no poles except at the origin).

FIR filters don't have poles inside the unit circle (other than trivial ones at the origin). This makes FIR filters **always stable**.

For low-pass filtering, place zeros on or near the **unit circle** in the **high-frequency region** (clustered around $z = -1$, or $\omega = \pi$)

For high-pass filtering, place zeros on or near the **unit circle** in the **low-frequency region** (clustered around $z = 1$, or $\omega = 0$)

FIR FILTER DESIGN USING POLE-ZERO

Question: Consider the following Filter Coefficients ($h[n]$)

$$h[0] = 0.5, h[1] = -0.5$$

The output $y[n]$ for an input $x[n]$ is calculated as:

$$y[n] = 0.5x[n] - 0.5x[n - 1]$$

Plot the pole-zero plot and state whether it is a high-pass or low-pass filter.

Ans.

$$H(z) = 0.5 - 0.5z^{-1} = \frac{z-1}{2z}$$

There is **one zero at** $z = 1$, which lies directly on the unit circle at $\omega = 0$ (low frequencies).

It is a High-Pass Filter.

EXERCISE FILTER DESIGN USING POLE-ZERO

Question: Consider the following Filter Coefficients ($h[n]$)

$$h[0] = 1, h[1] = 0.5$$

The output $y[n]$ for an input $x[n]$ is calculated as:

$$y[n] = x[n] + 0.5x[n - 1]$$

Plot the pole-zero plot and state whether it is a high-pass or low-pass filter.

IIR FILTER DESIGN USING POLE-ZERO

For an **Infinite Impulse Response (IIR)** filter, frequency shaping is controlled by **both zeros and poles**.

Unlike FIR filters (where poles are trapped at the origin and do nothing), IIR filters use feedback, placing poles at active locations on the z -plane.

Transfer function for IIR filter will contain both $Y(Z)$ and $X(Z)$.

Poles act like amplifiers or resonators—they **boost** frequencies near them, while zeros attenuate frequencies near them.

For high-pass filtering, place poles on or near the **unit circle** in the **high-frequency region** (clustered around $z = -1$, or $\omega = \pi$)

For low-pass filtering, place poles on or near the **unit circle** in the **low-frequency region** (clustered around $z = 1$, or $\omega = 0$).

Placement of zeroes will be same as FIR design.

IIR FILTER DESIGN USING POLE-ZERO

Example: Consider the following IIR filter

The output $y[n]$ for an input $x[n]$ is defined as:

$$y[n] = 0.5x[n] + 0.5x[n-1] + 0.5y[n-1]$$

Plot the pole-zero plot and state whether it is a high-pass or low-pass filter.

Ans.

$$H(z) = \frac{0.5 + 0.5z^{-1}}{1 - 0.5z^{-1}} = \frac{0.5(z+1)}{z-0.5}$$

POLE-ZERO MODELING FOR SPEECH

Speech can be modeled as:

$$s[n] = e[n] * h[n] \text{ where}$$

$e[n]$: excitation (voiced source like glottal pulses, or unvoiced noise) and

$h[n]$: vocal tract filter.

The vocal tract filter is represented by a **transfer function** with **poles** (resonances) and **zeros** (anti-resonances).

POLE-ZERO MODELING FOR SPEECH

Poles correspond to **formants** (resonant frequencies of the vocal tract).

Example: vowel /a/ has formants around 700 Hz, 1200 Hz, 2500 Hz.

These appear as poles near the unit circle in the z-plane.

Zeros model **anti-resonances**, such as nasal sounds or fricatives.

Example: nasal consonants introduce spectral dips due to zeros.

Sounds like /m/, /n/, /ng/ involve side cavities in the nasal tract that create sharp anti-resonances (zeros)

ONLINE RESOURCE REFERENCES

1. <https://www.youtube.com/watch?v=7Gl4kJUjp4c>
2. <https://www.youtube.com/watch?v=XJRW6jamUHk>
3. https://www.youtube.com/watch?v=xIN5Mnj_MAk